SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED
CO2: *Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analysing images.. (BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 3 **Total Marks:30**

Annexure A

Scenario-Based

Code of Conduct:

I Peram.Mahitha(192524136)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:*P.Mahitha*

RUBRICS FOR CALCULATION

Scenario based assessment						
Criteria	**Weight age (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Marks**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Annexure A

Scenario-Based

1. Crack Detection on Concrete Surfaces Using Drone Images

Scenario: A civil inspection team is developing a computer vision system to detect thin cracks on concrete bridges and building surfaces. Images are captured using a drone under outdoor conditions. Due to changing sunlight, shadows, and camera vibration, the raw images contain uneven illumination, low contrast, and Gaussian noise.
Questions:

A. Propose a suitable image preprocessing pipeline to prepare the drone images for crack detection. Your answer should include methods for illumination correction, noise reduction, contrast enhancement, and image normalization. Justify the need for each step.

B. Select an appropriate edge detection technique for identifying thin surface cracks. Compare it with at least one other edge detection method and explain why your selected technique is more suitable for detecting fine cracks.

C. After edge detection, some crack segments appear broken or disconnected. Suggest a suitable post-processing technique to connect the crack regions and improve the final detection output.

Ans)
Crack Detection on Concrete Surfaces Using Drone Images

A. Preprocessing pipeline

In this scenario, the drone images contain **uneven sunlight, shadows, low contrast, Gaussian noise and camera vibration**. Since the cracks are thin, directly applying edge detection may detect noise and shadows as cracks.

A suitable preprocessing pipeline is:

Drone image → Illumination correction → Gaussian filtering → Contrast enhancement → Normalization → Edge detection

1. Illumination correction

Because sunlight and shadows produce uneven brightness, use **background estimation or homomorphic filtering**.

This makes the illumination more uniform.
For example:

Bright region → corrected brightness
Dark shadow region → corrected brightness

This reduces false crack detection caused by shadows.

2. Noise reduction

The scenario specifies **Gaussian noise**, so a **Gaussian filter** is suitable.
A small 3 × 3 or 5 × 5 Gaussian filter can be applied.
It reduces random intensity variations while maintaining the overall crack structure.

3. Contrast enhancement

Use **CLAHE (Contrast Limited Adaptive Histogram Equalization)**.
It improves local contrast, making low-contrast cracks more visible against the concrete surface.

4. Image normalization

Normalize pixel values to a fixed range, such as 0 to 1.
Normalized value:

$$I_{norm} = \frac{I - I_{min}}{I_{max} - I_{min}}$$

This provides a consistent intensity range for images captured under different lighting conditions.

Final answer
The suitable preprocessing pipeline is:

Illumination correction → Gaussian noise reduction → CLAHE → Intensity normalization → Edge detection
This improves crack visibility, reduces noise and shadows, and prepares the image for accurate crack detection.

B. Suitable edge detection technique

For this scenario, **Canny edge detection** is the most suitable.
The cracks are thin and may have low contrast. Canny is useful because it provides good edge localization and detects weak edges.

Canny works as:

Gaussian smoothing → Gradient calculation → Non-maximum suppression → Double threshold → Hysteresis
Comparison with Sobel

Feature	**Sobel**	**Canny**
Noise handling	Moderate	Better
Thin crack detection	Good	Very good
Edge localization	Moderate	High
Weak edges	Less effective	More effective
Computation	Low	Higher

For drone-based crack detection, **Canny is preferred** because it produces thin, well-localized edges and can preserve weak crack segments.

C. Connecting broken crack segments

After Canny edge detection, some cracks may appear broken because of noise, low contrast or shadows.

Use **morphological closing**.

Closing consists of:

$$\text{Closing} = \text{Dilation} + \text{Erosion}$$

First, dilation expands the detected crack pixels and connects small gaps.

Then, erosion reduces the expanded region while maintaining the connected structure.

Therefore:

Canny edges → Dilation → Erosion → Connected crack regions

A small structuring element should be used so that separate cracks are not incorrectly joined.

2. Human Motion Tracking in a Security Camera Video

Scenario: A security camera is installed in a hallway to monitor people moving in different directions. The system must estimate the motion of each person across video frames for tracking and activity analysis. The hallway contains plain walls, smooth floors, and occasional shadows.

Questions:

A. Explain how the Lucas-Kanade optical flow method can be used to estimate the motion of people between consecutive video frames.

B. Discuss how the aperture effect can create errors in motion estimation, especially on uniform regions such as plain walls, floors, and clothing surfaces.

C. If a person suddenly changes speed or direction, describe how the optical flow vectors will be affected. Suggest a method to improve tracking accuracy in such situations.

Ans)

Human Motion Tracking in a Security Camera Video

A. Lucas-Kanade optical flow

In this scenario, a security camera captures people walking in different directions through a hallway.

The **Lucas-Kanade optical flow method** estimates how image points move between two consecutive frames.

Suppose a person appears at:

Frame 1: (x_1, y_1)

Frame 2: (x_2, y_2)

The displacement is:

$$u = x_2 - x_1$$
$$v = y_2 - y_1$$

The optical flow vector is:

$$V = (u, v)$$

The Lucas-Kanade method uses the optical-flow equation:

$$I_x u + I_y v + I_t = 0$$

It uses several nearby pixels in a small window to estimate $u$and v.

Example

Suppose a person's feature moves from:

$$(100, 150) \rightarrow (106, 154)$$

Then:

$$u = 106 - 100 = 6$$
$$v = 154 - 150 = 4$$

Therefore:

$$V = (6, 4)$$

Magnitude:

$$|\, V \,| = \sqrt{6^2 + 4^2}$$
$$= \sqrt{36 + 16}$$
$$= \sqrt{52}$$

$$\approx 7.21 \text{ pixels/frame}$$

Direction:

$$\theta = \tan^{-1}\left(\frac{4}{6}\right)$$
$$\theta \approx 33.69°$$

Thus, the person is moving approximately 7.21 pixels/frame at an angle of 33.69°.

B. Aperture effect

The hallway contains **plain walls, smooth floors and clothing surfaces**.

These regions have very little texture.

For example, if a person is wearing a plain shirt, many nearby pixels have almost the same intensity.

The optical flow method cannot easily determine the exact direction of motion from such a uniform region.

This is called the **aperture effect**.

For example:

Uniform region → insufficient intensity variation → ambiguous motion → incorrect flow vector

For a straight edge, the system can determine motion perpendicular to the edge but cannot reliably determine motion along the edge.

Therefore, plain walls, floors and clothing can cause inaccurate motion estimation.

C. Sudden change in speed or direction

Suppose a person is initially moving to the right and suddenly turns left.

Before turning:

$$V_1 = (5, 0)$$

After turning:

$$V_2 = (-, 41)$$

The direction and magnitude of the optical flow vectors change suddenly.

This may cause:

- Incorrect feature matching
- Tracking errors
- Lost feature points
- Wrong motion prediction

Lucas-Kanade normally works best when motion between consecutive frames is relatively small.

Improvement

Use **Pyramidal Lucas-Kanade optical flow** to handle larger movements.

A **Kalman filter** can also be used to predict the person's next position.

Therefore:

Pyramidal Lucas-Kanade + Kalman filter → improved tracking during sudden motion changes.

3.Vehicle Tracking at a Traffic Junction Using Video Frames

Scenario: A smart traffic monitoring system uses CCTV footage to track vehicles at a busy road junction. Vehicles move at different speeds, some stop suddenly at signals, and others turn left or right. The video also contains shadows, reflections, and partial occlusion when vehicles overlap.

Questions:

A. Explain how optical flow can be used to estimate the direction and speed of moving vehicles in the video sequence.

B. Identify two challenges that may affect accurate motion estimation in this scenario, such as occlusion, shadows, sudden braking, or camera vibration. Explain how each challenge affects the optical flow result.

C. Suggest a suitable improvement method to make vehicle tracking more reliable when vehicles overlap or change direction suddenly.

Ans)

Vehicle Tracking at a Traffic Junction Using Video Frames

A. Optical flow for vehicle direction and speed

In this scenario, vehicles move at different speeds and may stop, turn left or turn right.

Optical flow compares consecutive video frames and calculates the movement of vehicle pixels.

The optical flow vector is:

$$V = (u, v)$$

where:

- u= horizontal displacement
- v= vertical displacement

The motion magnitude is:

$$M = \sqrt{u^2 + v^2}$$

The direction is:

$$\theta = \tan^{-1}\left(\frac{v}{u}\right)$$

Example

Suppose a vehicle moves:

$$u = 8,\ v = 6$$

Magnitude:

$$\begin{aligned} M &= \sqrt{8^2 + 6^2} \\ &= \sqrt{64 + 36} \\ &= \sqrt{100} \\ &= 10 \text{ pixels/frame} \end{aligned}$$

Direction:

$$\theta = \tan^{-1}\left(\frac{6}{8}\right)$$
$$\theta \approx 36.87^\circ$$

Therefore, the vehicle moves at approximately **10 pixels/frame** with a direction of **36.87°**.

If the time between frames is Δt, then image-plane speed is:

$$Speed = \frac{M}{\Delta t}$$

A stationary vehicle will have an optical-flow magnitude close to zero.

B. Two challenges affecting motion estimation

1. Occlusion

At a busy junction, two vehicles may overlap.

Suppose Vehicle A is behind Vehicle B.

When B covers A, the features of A disappear temporarily.

Therefore:

Vehicle overlap → Missing features → Incorrect optical flow → Tracking error

The system may even assign Vehicle A's movement to Vehicle B.

This makes vehicle tracking difficult.

2. Shadows

Vehicles create shadows on the road.

When the vehicle moves, its shadow also moves.

Optical flow may interpret the moving shadow as part of the vehicle.

Therefore:

Moving shadow → False motion vectors → Incorrect vehicle boundary

This can cause inaccurate vehicle detection and tracking.

C. Improvement for overlapping and sudden direction changes

A better approach is to combine:

Object detection + Optical flow + Kalman filter

Step 1: Object detection

Detect each vehicle separately and assign an ID.

For example:

Vehicle1
Vehicle2
Vehicle 3

Step 2: Optical flow

Calculate the movement of features inside each detected vehicle.

Step 3: Kalman filter

Predict the next vehicle position using its previous position and velocity.

For example:

Previous position = 100 pixels

Velocity = 10 pixels/frame

Predicted next position:

$$100 + 10 = 110 \text{ pixels}$$

If the vehicle is temporarily hidden because of occlusion, the predicted position helps maintain its track.

Step 4: Handle direction changes

When a vehicle suddenly turns, new optical-flow vectors are calculated and the Kalman filter updates its predicted direction.

Final system

CCTV video → Vehicle detection → Optical flow → Kalman prediction → Occlusion handling → Vehicle tracking

This combined method is more reliable than using optical flow alone because it can maintain vehicle identities during overlap, sudden braking and sudden changes in direction.